FINAL REPORT · PUERTO VALLARTA · SUNDAY 11 OCTOBER 2026, 3 PM

Hurricane Simon and Puerto Vallarta: what the models said, what happened, and how much to trust the next 48 hours

A verification of 25 forecast models (physics, AI, ensembles and the National Hurricane Center) against the observed storm, with the uncertainty shown rather than hidden, and a point-by-point reply to the uploaded analysis *Comprehensive Verification and Theoretical Analysis of Dynamical and AI-Driven Tropical Cyclone Forecasting*.

Based on NHC advisory 18 (3 pm CST) · SMN aviso 34 (12:00 CST) · ATCF decks · ECMWF open data · NOAA open data (GFS, GEFS, AIGFS, NBM, HAFS) · NOAA/CIRA AI-model archive · METAR MMPR · GOES-18

> **Not an official forecast.** Follow the SMN, the NHC and Protección Civil. Emergencies: 911. Everything here is an independent reading of public data; it is meant to help you interpret the official products, not replace them.

1. The short version

- **Where Simon is.** At 3 pm Sunday the center was at 19.6N 105.8W, 90 km south of Cabo Corrientes, with 130 mph (115 kt) winds, moving NNW at 9 km/h, roughly parallel to the coast. The eye is brushing the Jalisco coast but is not forecast to cross it in the next two days.
- **Which models were right.** Over the last 48 hours the best position forecasts came from Google DeepMind's cyclone model (mean error 33 km; 5 km off at 3 pm today), the NHC's corrected consensus (38 km) and the NHC official forecast (47 km). The least accurate of the fully verified models were the US regional hurricane models (HAFS-B, COAMPS-TC, HAFS-A, HMON) and UKMET; the GFS-driven AI runs (Aurora, FourCastNet, Pangu) did as badly or worse at the leads they could be checked. From Saturday evening HAFS-A, HAFS-B and UKMET carried Simon about 150 km too far north, into or past Banderas Bay, and HMON and COAMPS-TC had done the same (140–280 km) in their Saturday daytime runs. That is where the hurricane-wind scenario for Vallarta came from.
- **Intensity.** Every gridded AI model collapsed Simon to a tropical storm within 12–24 hours while it stayed a category 4; so did the physics ensembles on their 25-km public grids. The best intensity guidance for this storm was the NHC itself, its corrected consensus, the statistical SHIPS model and DeepMind's cyclone model.
- **Vallarta so far.** The airport's reports show a peak wind of 17 km/h (9 kt) in the last 48 hours, with a 19-hour gap in reports from Saturday 6 pm to Sunday 1 pm, and about 9 mm (0.35 in) of rain from one shower in the six hours to 1:44 pm. Residents in the city, including the person this report is for, saw no rain through early

afternoon; there is no official gauge in town. Of the seven gridded models that give a rain total, five gave Vallarta more rain than fell; ECMWF AIFS (7 mm) and AIGFS (10 mm) matched. Almost all gridded models gave less wind than measured; the track-based estimates gave more.

- **What to expect.** The weighted consensus puts the closest pass about 100–110 km west-southwest of the city around 11:30 pm tonight, with the center still a hurricane. For the city: gusty wind from the east and east-southeast, sustained roughly 30–45 km/h with gusts 45–60 km/h on high floors, from Sunday afternoon to about 4 am Monday, peaking around 7–8 pm and staying near that level until the center passes around midnight; little rain tonight; a chance of moderate rain Monday afternoon when the wind turns onshore; a dangerous sea on the outer coast (SMN: waves of 8 to 10 m and a surge of 1.5 to 2.5 m on the Jalisco coast).
- **The uncertainty, in one line.** The NHC puts Vallarta's chance of tropical-storm-force wind at 8% and of hurricane-force wind under 1%; of the models that have earned trust in this storm, none brings the center within 90 km of the city. A 30–40 km shift east of the line, inside the two-day average error of most models and at the edge of the best three's (33–47 km), would bring tropical-storm-force gusts over the bay tonight, with sustained winds just short of it.
- **On the uploaded analysis.** Its physics and its description of the AI intensity problem hold up. Its model-verification table does not: the numbers have no source, and several contradict the official record (HAFS-B was the worst track model of the last two days, not a 51-nm one; no dynamical model showed "absolute supremacy" on intensity). It also omits the one model that actually won, and the 15 inches of rain it uses as an example for Vallarta is a hypothetical, not a forecast: the city has had under 10 mm so far.

2. Where Simon is and what is forecast

NHC advisory 18 (3 pm CST): center 19.6N 105.8W, 115 kt, 947 mb, moving 335° at 5 kt. Forecast: 20.2N 106.1W and 90 kt by midnight tonight (near the coast), 21.0N 106.6W and 80 kt by noon Monday, 21.6N 106.9W and 75 kt by midnight Monday, then a bend back toward the Nayarit–Sinaloa coast by Wednesday, still a 75-kt hurricane in the NHC forecast (a Hurricane Watch is in effect from north of El Roblito to Bahía Tempehuaya), weakening inland on Thursday. The discussion calls the track forecast "low confidence" and notes the guidance has shifted west and slower again. The wind-speed probabilities for Puerto Vallarta are 8% for tropical-storm force and under 1% for hurricane force.

SMN aviso 34 (12:00 CST) keeps extraordinary point rainfall (more than 250 mm) for coastal and western Jalisco, 150–250 mm in Colima, 75–150 mm in southern Nayarit, waves of 8 to 10 m on the Jalisco coast, a storm surge of 1.5 to 2.5 m on the Jalisco and Colima coasts, and gusts of 240–250 km/h on the Jalisco coast near the center, decreasing through the day. The SMN's wording is "Jalisco (coast and west)", which on paper includes the city; the NHC's "isolated maximum storm totals of 30 inches" are for "portions of the states of Michoacán, Colima and Jalisco". My reading, not the agencies': with the center to the southwest, the heaviest totals fall on the windward sierra south and west of the bay, while the city sits on the lee side in descending easterly flow, as today's dry afternoon showed. That reading changes once the wind turns onshore.



Figure 1. The one-page summary for the three places, from the advisory-18 data: timing, peak conditions and the worst plausible impact at each. The wind numbers are the NHC wind field on the NHC line at the city, capped by the wind value the NHC odds give a 50% chance of (the cap is not reached today), scaled to each building's exposure; the rain from the GOES-18 satellite rain estimate carried along the NHC line, never scaled up to the SMN regional figure. Spanish version on the shared page.

3. The uploaded analysis: what it argues, and where the data agrees

The document is a long essay in five parts: the physics of tropical cyclones and of this coastline; the architecture of AI weather models and why they under-forecast intensity (training on 31-km reanalysis, a mean-squared-error loss that rewards smoothing, the "double penalty"); pathways to fix that (high-resolution training, physics-informed networks, neural ODEs, and ECMWF's AIFS-TC post-processing); a model-verification table for Simon; and a critique of the NHC cone of uncertainty as a communication device. Much of it is sound. The parts that matter for someone deciding what to do tonight are the verification table and the rainfall claim, and those are the weakest parts. The table below checks each claim against the record we built from the ATCF decks, the ECMWF and NOAA open-data feeds and the airport observations.

Claim in the document	What the record shows	Verdict
"Pure AI models (GraphCast, AIFS, Pangu-Weather) vastly outperformed the traditional dynamical models (GFS, ECMWF) in predicting the exact … positioning of Hurricane Simon."	Partly. Over the last 48 h, ECMWF AIFS averaged 50 km against 69 km for ECMWF IFS and 67 km for GFS. But GraphCast (66 km, incomplete) and Pangu (107 km, incomplete), run by NOAA/CIRA from GFS analyses, were no better than GFS. The best position model was DeepMind's cyclone model at 33 km, which the document never mentions, and the NHC's corrected consensus (38 km) and official forecast (47 km) beat every gridded AI model.	**Overstated**
"The dynamical models exhibited absolute supremacy regarding intensity prediction." HAFS-B "~7.8 kt" mean intensity error; HAFS-A ~9.5 kt.	No. Whole-storm mean intensity errors at 12–36 h: NHC official 5.6 kt, corrected consensus 7.7, statistical DSHP 13.0, DeepMind cyclone model 14.6, HAFS-A 18.0, HMON 20.0, HWRF 20.8, HAFS-B 20.8. From the Saturday 6 pm run, HAFS-A and HAFS-B forecast 34 and 32 kt for 6 pm Sunday, against 115 kt observed at 3 pm: they weakened it because they took it inland about 150 km too far north (Figure 4).	**Contradicted**
Verification table: GraphCast ~39.5 nm, AIFS ~38.0, Pangu ~41.2, HAFS-A ~48.2, HAFS-B ~51.0, ECMWF HRES ~58.4, GFS ~62.1 nm track error for "72-hour lead time forecasts (initialized on October 8, 2026)".	No source, no cycle hours and no averaging method are given. Computing that metric from the public decks (the four Oct 8 cycles, 72-h forecasts against the best track) gives HAFS-A 87 nm, HAFS-B 83 nm, HWRF 56 nm, GFS 106 nm and the NHC 70 nm for track, with intensity errors from 36 kt (NHC) to 64 kt (HAFS-B): every model, physics or AI, missed the rapid intensification at 72 h by 25–110 kt, so the table's 8–10 kt for the HAFS models cannot be right. In our 48-hour window HAFS-B had the largest errors of any model (169 km mean; 149 km too far north at 3 pm) and HAFS-A 117 km.	**Unsupported**
"Uncorrected AI models forecasted Simon as a mere Category 1 or strong Tropical Storm, missing the rapid intensification event entirely."	Confirmed. For 6 pm Sunday, from the Saturday 6 pm analysis (115 kt observed at 3 pm): ECMWF AIFS 34 kt, FourCastNet 13, Pangu 18, Aurora 18, AIGFS 16, GraphCast 24 kt at 18 h. The physics ensembles on their 25-km public grids did the same: the median ECMWF ensemble member had 50 kt at 24 h (members 20–106), the AIFS ensemble median 33 kt.	**Confirmed**

Claim in the document	What the record shows	Verdict
The cause is resolution of the training data plus the smoothing effect of the MSE loss ("double penalty").	Consistent with what we saw, and the standard explanation. One addition: for GFS, AIGFS and the CIRA runs part of the collapse came from drifting the low inland, so track and intensity errors compound; the DeepMind cyclone model, trained for cyclones specifically, kept 91 kt at 24 h.	**Reasonable**
AIFS-TC post-processing cuts the AI intensity bias "from nearly -29 knots to an operational margin of about -2 knots" and "was autonomously designed and built by a Large Language Model coding agent".	Not testable here. AIFS-TC output is not part of the ECMWF open-data feed we used, and the claims about its construction come from the cited preprint. They are reported, not verified.	**Unverified**
"When the storm's track shifts, or when a massive, highly asymmetric system like Simon brings 15 inches of rain to a mountainous area like Puerto Vallarta that was technically 'outside' the center-track cone, it breeds profound public distrust."	Used as an example, not a forecast, but it has not happened: by 3 pm Sunday residents reported no rain in the city and the airport about 9 mm. The SMN's "more than 250 mm" is for "Jalisco (coast and west)" and the NHC's "isolated maximum storm totals of 30 inches" for "portions of the states of Michoacán, Colima and Jalisco", in practice the windward terrain; with the center to its southwest, Vallarta is in descending easterly flow.	**Illustrative, overstated**
The Sierra Madre shields the city from the strongest wind; orographic lift makes flash flooding and mudslides the lethal hazard.	Agree on the mechanism. Our bay-wide fields show the same: wind reaching the city from the east and east-southeast, rain concentrated on the sierra, an onshore turn only once the center is northwest of the bay (Monday afternoon in the consensus).	**Agree**
Rapid intensification driven by deep ocean heat content and low shear.	Consistent. The best track shows 65 kt at 6 pm Thursday and 105 kt at 6 pm Friday (+40 kt in 24 h), then 135 kt by Saturday evening.	**Consistent**
The cone of uncertainty is misread (size, center-line and edge fallacies) and triggers anxiety; the 2026 cone embeds inland watches and warnings and experimental versions have sought to capture 90% of errors.	Agree on the substance; the three fallacies are well documented, and the inland watches-and-warnings cone is real. The specific "90%" figure and the psychometric claims could not be checked from here. See section 7 for what we did instead.	**Agree, specifics unverified**

The document's 60-item reference list is not reproduced here. Several of its entries are forum threads and news items; the technical claims that matter are checked above against primary data rather than against those sources.

4. The verification, model by model

Every model was scored the same way. **Position:** the error of each forecast made in the 48 hours before 3 pm Sunday, measured against the NHC's official positions every six hours and against the advisory-18 position; the ranking score is the mean at 12, 24 and 36 h lead. **Vallarta wind and rain:** what the run of Saturday 6 pm gave the city up to 3 pm Sunday, against the airport record (max 9 kt in the available reports; about 9 mm of rain by 1:44 pm). Gridded models were sampled at their nearest grid point to the city; track models give no point wind, so their wind is the value implied by the NHC wind structure, or the model's own wind radii where the deck carries them, placed on the model's line.

What stands out

- **The best three were not the newest or the highest-resolution models.** DeepMind's cyclone model, the NHC's corrected consensus and the NHC official forecast. The simple climatology-and-persistence baseline (53 km) beat GFS, GEM, UKMET and all five US regional hurricane models (HWRF, HMON, HAFS-A, HAFS-B, COAMPS-TC) over this window, which is unusual and says the storm's motion was steady while many models were looking for a turn.
- **The regional hurricane models failed in the direction that mattered.** HAFS-A, HAFS-B, HMON and COAMPS-TC were 150–280 km too far north at various cycles, bringing a category-4 core into Banderas Bay. Because these are the models the NHC leans on for structure, that is also why the official forecast was 60–140 km too far east and north on Friday and Saturday, and why the NHC's own probability of hurricane-force wind in Vallarta reached 15–20% late Friday and early Saturday (section 6).
- **Among gridded AI models, ECMWF AIFS and its ensemble did well on position** (50 and 58 km; 43 of 51 ensemble members within 100 km of the true center at 3 pm). The GFS-initialised models in the NOAA/CIRA archive did not: FourCastNet and Aurora were 160–200 km off at 3 pm, and the NOAA AIGFS dissipated the storm inland and ended 210 km too far north.
- **Local wind and rain are beyond all of them.** Nine gridded models put the city's peak wind at 4–12 km/h against 17 km/h measured, and five of the seven that carry rain gave more than fell. The bay, the city and the sierra do not exist on a 25–50 km grid; the NBM, the US blend that corrects for terrain, reaches Vallarta only through its oceanic domain (wind only); its 36 km/h was above the airport's 17 km/h, as were the track-based estimates (28–75 km/h), while the coarse grids were below it.

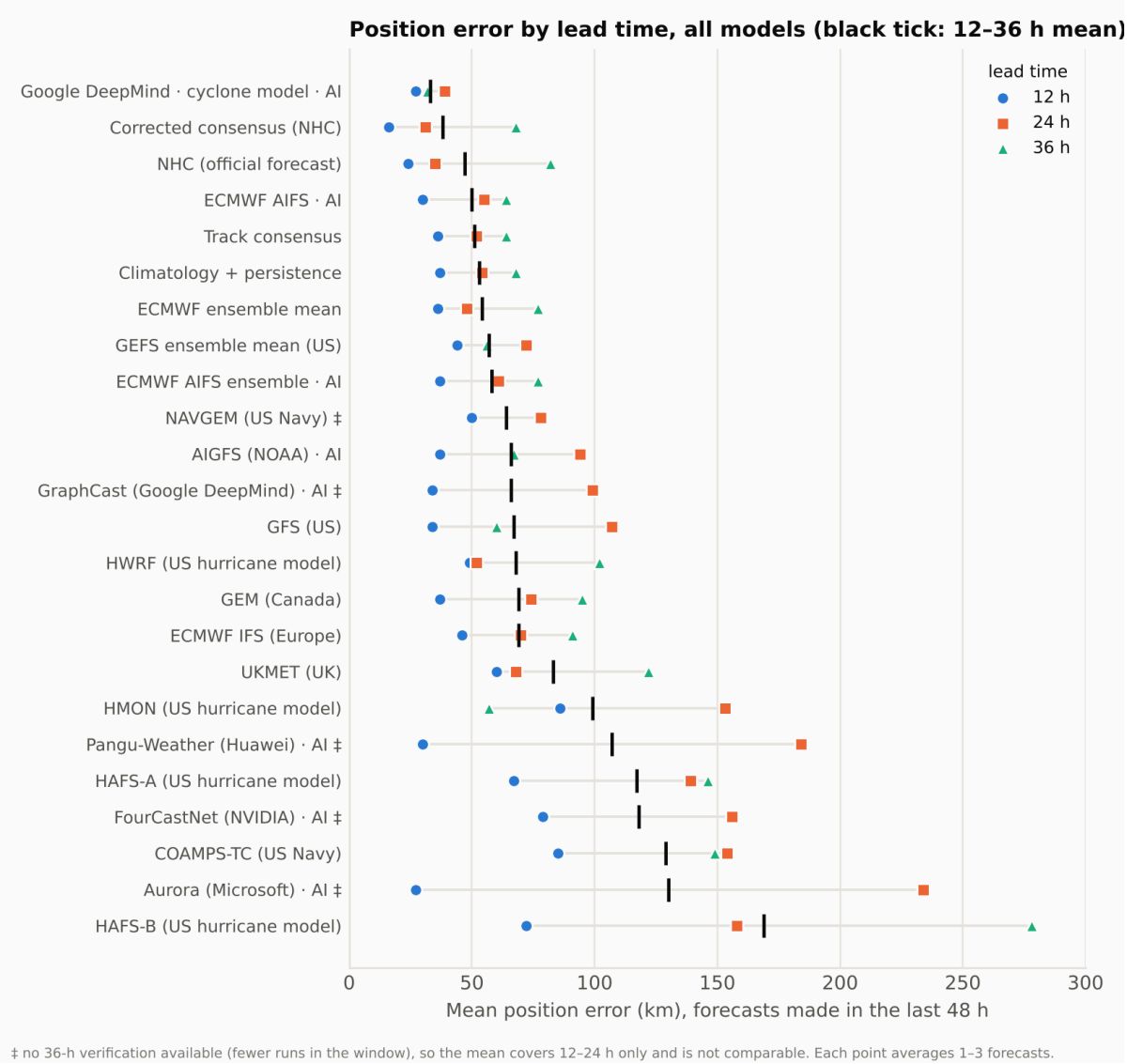


Figure 2. Position error by lead time. The black tick is the 12–36 h mean used to rank the models. Models marked ‡ lack a 36-h verification inside the window, so their score covers 12–24 h only and is not directly comparable. Each point averages one to three forecasts: a small sample, so the order of neighbours can change, but the gap between the top three and the bottom five is far larger than the noise.

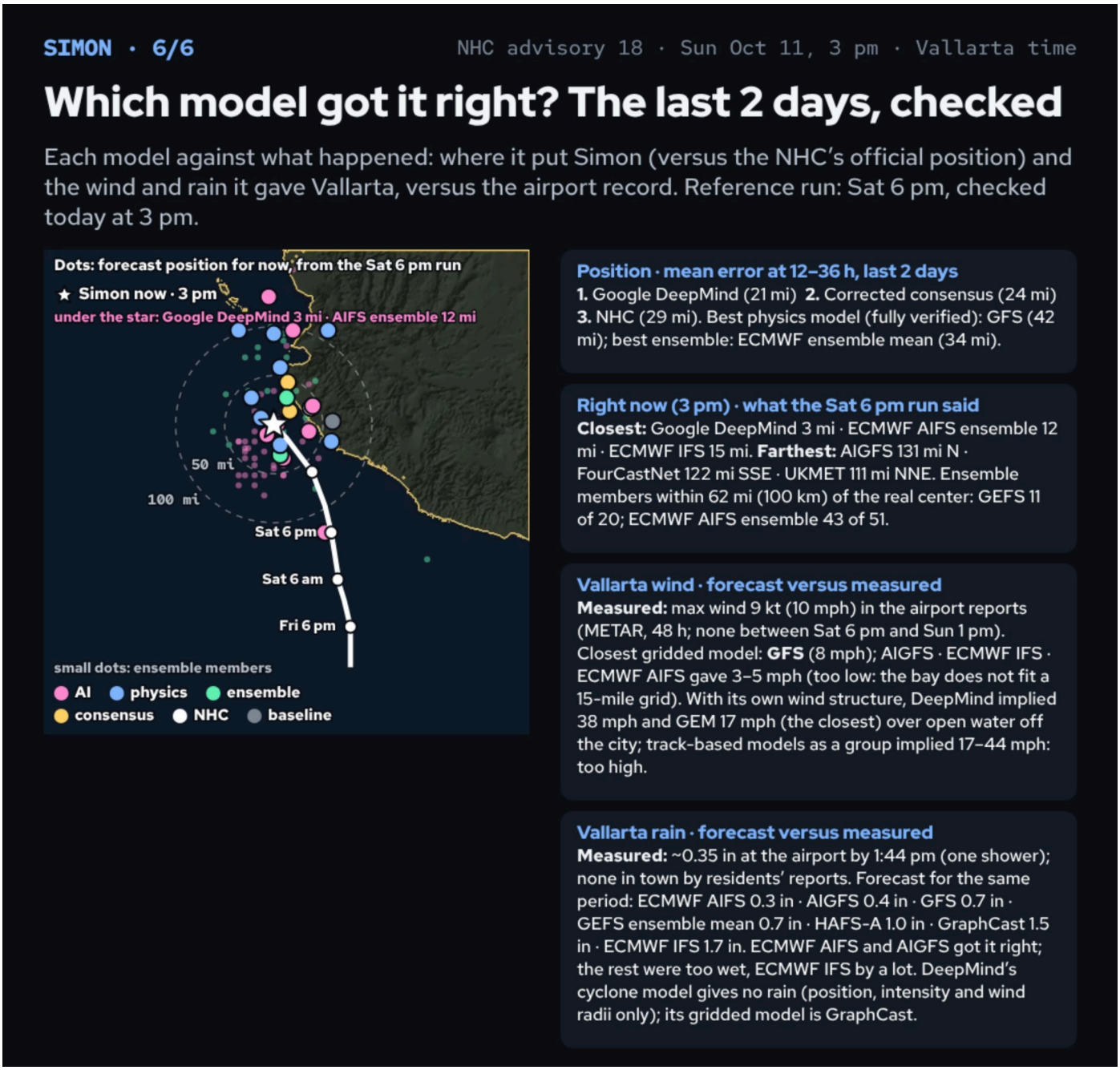
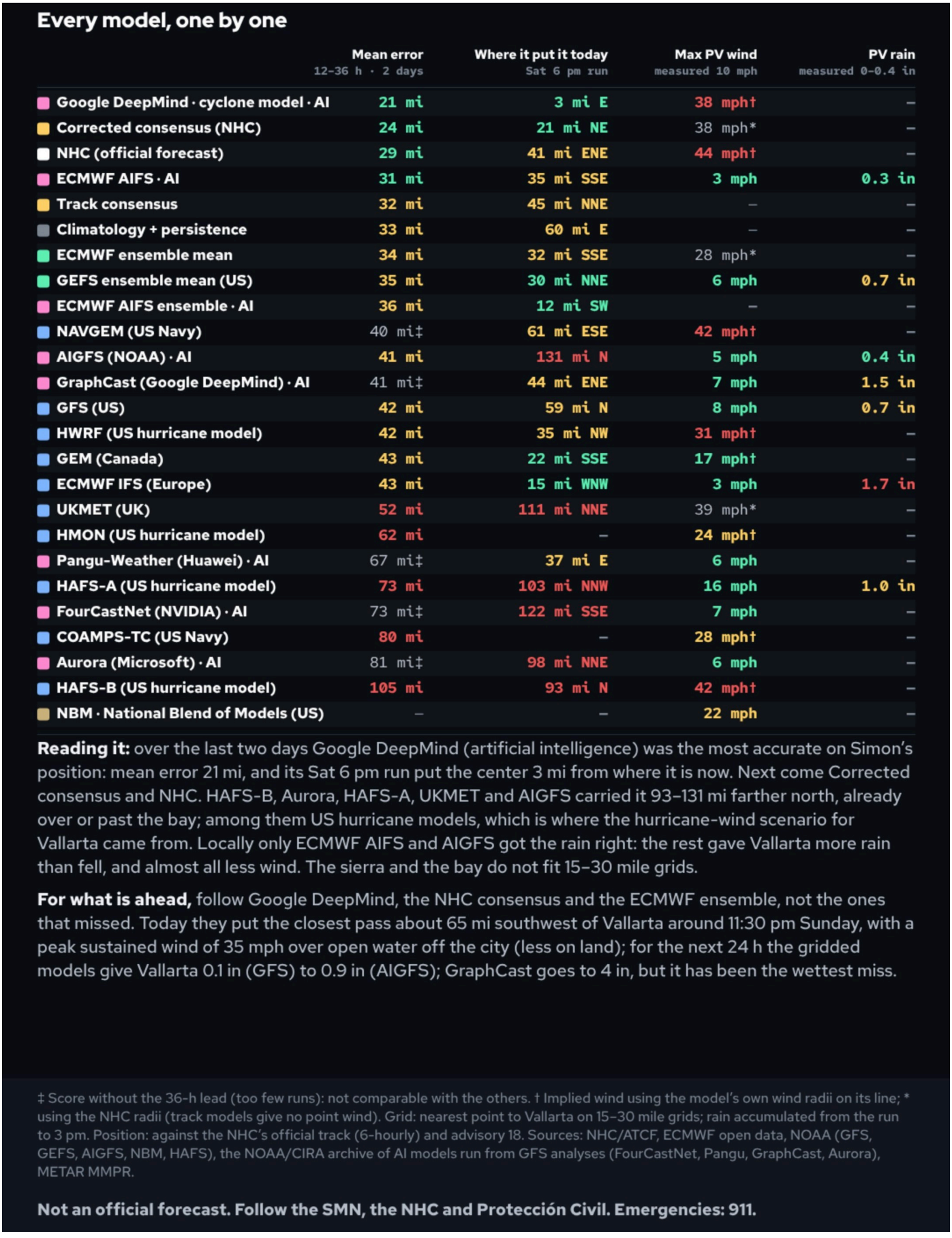


Figure 3a. The verification graphic shared with residents (English; Spanish on the shared page), upper part. Map: where each Saturday-evening run put Simon for 3 pm Sunday, with the ensemble members as small dots.



Every model, one by one

	Mean error 12–36 h · 2 days	Where it put it today Sat 6 pm run	Max PV wind measured 10 mph	PV rain measured 0–0.4 in
Google DeepMind · cyclone model · AI	21 mi	3 mi E	38 mph†	–
Corrected consensus (NHC)	24 mi	21 mi NE	38 mph*	–
NHC (official forecast)	29 mi	41 mi ENE	44 mph†	–
ECMWF AIFS · AI	31 mi	35 mi SSE	3 mph	0.3 in
Track consensus	32 mi	45 mi NNE	–	–
Climatology + persistence	33 mi	60 mi E	–	–
ECMWF ensemble mean	34 mi	32 mi SSE	28 mph*	–
GEFS ensemble mean (US)	35 mi	30 mi NNE	6 mph	0.7 in
ECMWF AIFS ensemble · AI	36 mi	12 mi SW	–	–
NAVGEM (US Navy)	40 mi‡	61 mi ESE	42 mph†	–
AIGFS (NOAA) · AI	41 mi	131 mi N	5 mph	0.4 in
GraphCast (Google DeepMind) · AI	41 mi‡	44 mi ENE	7 mph	1.5 in
GFS (US)	42 mi	59 mi N	8 mph	0.7 in
HWRF (US hurricane model)	42 mi	35 mi NW	31 mph†	–
GEM (Canada)	43 mi	22 mi SSE	17 mph†	–
ECMWF IFS (Europe)	43 mi	15 mi WNW	3 mph	1.7 in
UKMET (UK)	52 mi	111 mi NNE	39 mph*	–
HMON (US hurricane model)	62 mi	–	24 mph†	–
Pangu-Weather (Huawei) · AI	67 mi‡	37 mi E	6 mph	–
HAFS-A (US hurricane model)	73 mi	103 mi NNW	16 mph	1.0 in
FourCastNet (NVIDIA) · AI	73 mi‡	122 mi SSE	7 mph	–
COAMPS-TC (US Navy)	80 mi	–	28 mph†	–
Aurora (Microsoft) · AI	81 mi‡	98 mi NNE	6 mph	–
HAFS-B (US hurricane model)	105 mi	93 mi N	42 mph†	–
NBM · National Blend of Models (US)	–	–	22 mph	–

Reading it: over the last two days Google DeepMind (artificial intelligence) was the most accurate on Simon's position: mean error 21 mi, and its Sat 6 pm run put the center 3 mi from where it is now. Next come Corrected consensus and NHC. HAFS-B, Aurora, HAFS-A, UKMET and AIGFS carried it 93–131 mi farther north, already over or past the bay; among them US hurricane models, which is where the hurricane-wind scenario for Vallarta came from. Locally only ECMWF AIFS and AIGFS got the rain right: the rest gave Vallarta more rain than fell, and almost all less wind. The sierra and the bay do not fit 15–30 mile grids.

For what is ahead, follow Google DeepMind, the NHC consensus and the ECMWF ensemble, not the ones that missed. Today they put the closest pass about 65 mi southwest of Vallarta around 11:30 pm Sunday, with a peak sustained wind of 35 mph over open water off the city (less on land); for the next 24 h the gridded models give Vallarta 0.1 in (GFS) to 0.9 in (AIGFS); GraphCast goes to 4 in, but it has been the wettest miss.

‡ Score without the 36-h lead (too few runs): not comparable with the others. † Implied wind using the model's own wind radii on its line; * using the NHC radii (track models give no point wind). Grid: nearest point to Vallarta on 15–30 mile grids; rain accumulated from the run to 3 pm. Position: against the NHC's official track (6-hourly) and advisory 18. Sources: NHC/ATCF, ECMWF open data, NOAA (GFS, GEFS, AIGFS, NBM, HAFS), the NOAA/CIRA archive of AI models run from GFS analyses (FourCastNet, Pangu, GraphCast, Aurora), METAR MMPR.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

Figure 3b. The same graphic, lower part: all 25 models, with the reading and the notes.

5. The intensity problem, shown on Simon

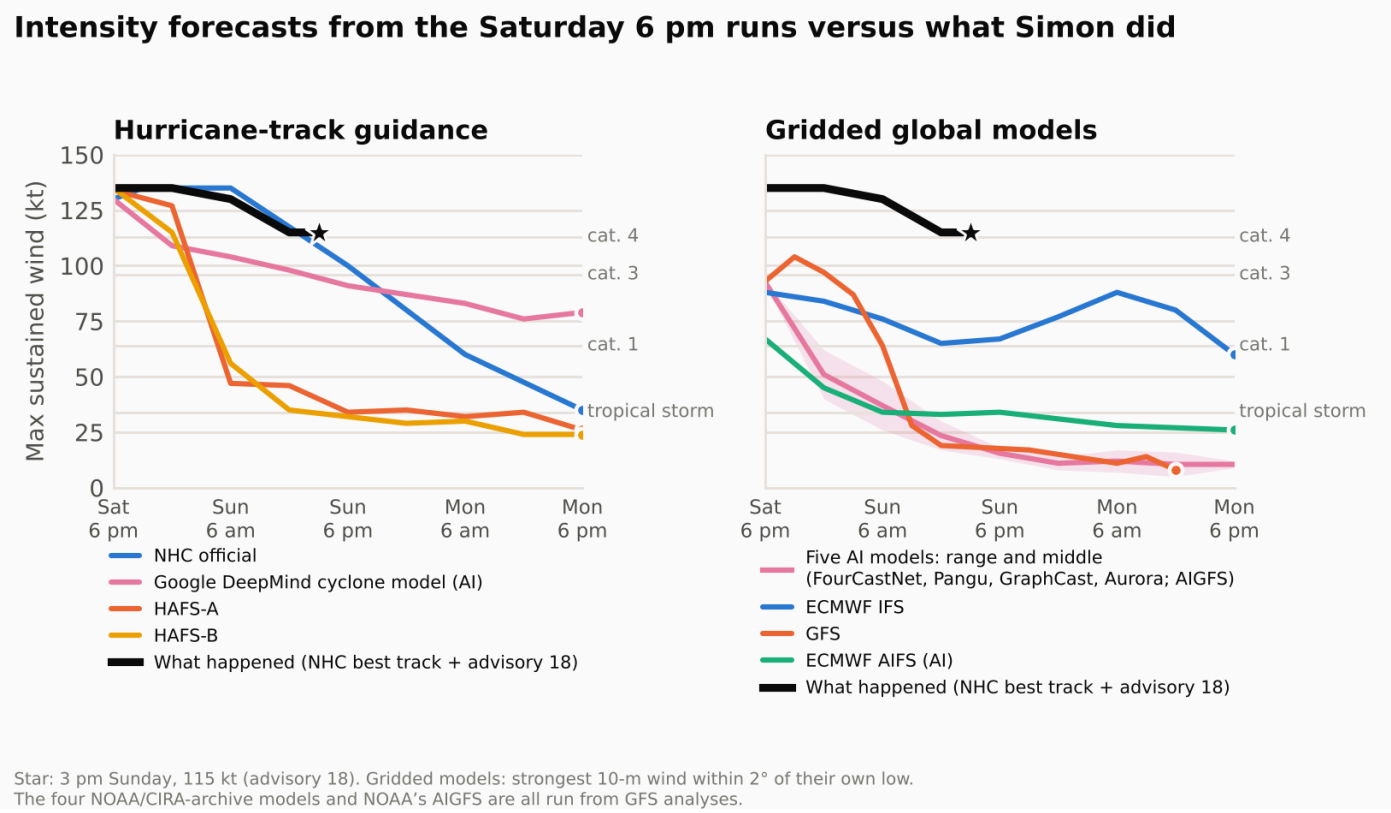


Figure 4. Intensity forecasts from the Saturday 6 pm runs. Left: the guidance the NHC uses. Right: gridded global models, with the four AI models from the NOAA/CIRA archive and NOAA's AIGFS as a band. The black line is what Simon did; the star is advisory 18.

Three different failures are visible in one picture. The **gridded AI models** (right, pink band and the AIFS line) start from the same 93-kt GFS or 67-kt AIFS analysis and lose half their wind in the first six hours, then settle at 15–35 kt: the smoothing the document describes, plus a 0.25° grid that cannot hold a 20-km eyewall. The **regional hurricane models** (left, HAFS-A and HAFS-B) start at the right intensity and crash to about 35 kt by 18–24 h, not because of smoothing but because they ran the storm into the Sierra Madre 150 km north of where it went. The **global physics models** on their 25-km public grids (ECMWF IFS, GFS) sit in between: IFS kept 65–88 kt, GFS fell to 19 kt as it too took the center inland.

The guidance that worked was the guidance that treats intensity as its own problem: the NHC forecaster (135 kt at 12 h, 100 kt at 24 h), the corrected consensus and the DeepMind cyclone model (104 kt at 12 h, 91 kt at 24 h), which is trained on cyclone tracks and intensities rather than on gridded fields. That supports one half of the document's prescription, a cyclone-specific correction on top of the global model (what AIFS-TC and the DeepMind model are), and does not test the other half, physics-informed networks.

Two caveats. The intensity verification covers one storm and a handful of cycles. And "right" here means right for Vallarta's purposes, where the question was whether a category-4 core would reach the bay; for the coast between Playa Pérula and Cabo Corrientes, which the eastern eyewall is brushing this evening, the intensity the NHC carried was the relevant number, and it was right.

6. Uncertainty, shown rather than hidden

The document's criticism of the cone is that it looks certain. The honest alternative is to show how the forecast itself moved, how wide the ensembles are, and what the models' own error record implies for tonight.

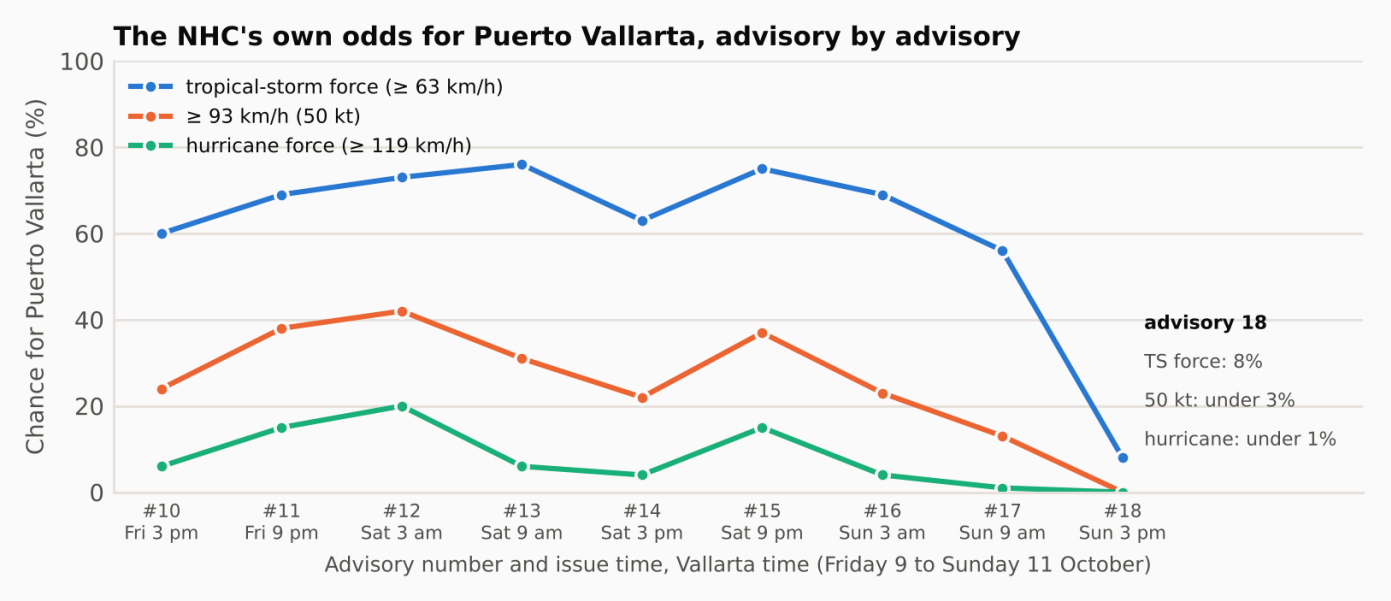


Figure 5. The NHC's cumulative probability of tropical-storm-force, 50-kt and hurricane-force wind at Puerto Vallarta, advisory by advisory. These numbers were always available in the wind-speed probability product; the public graphics showed the cone instead.

From Friday afternoon to Saturday night (advisories 10 to 15) the chance of hurricane-force wind in the city moved between 4% and 20%, and the chance of tropical-storm-force wind between 60% and 76%. That is the period in which the regional hurricane models carried the core into the bay. Since then it has gone 75%, 69%, 56% and, at 3 pm today, 8%: almost the whole drop came in one advisory, between 9 am and 3 pm, as the storm held its north-northwest heading and the guidance shifted west. A number that can fall 48 points in six hours can also climb back. Anyone who read the probabilities rather than the cone saw a "likely tropical-storm wind, unlikely hurricane" picture on Saturday and a "probably neither" picture today. The cone, drawn around the official line that was 60–140 km too far east, looked the same both days.

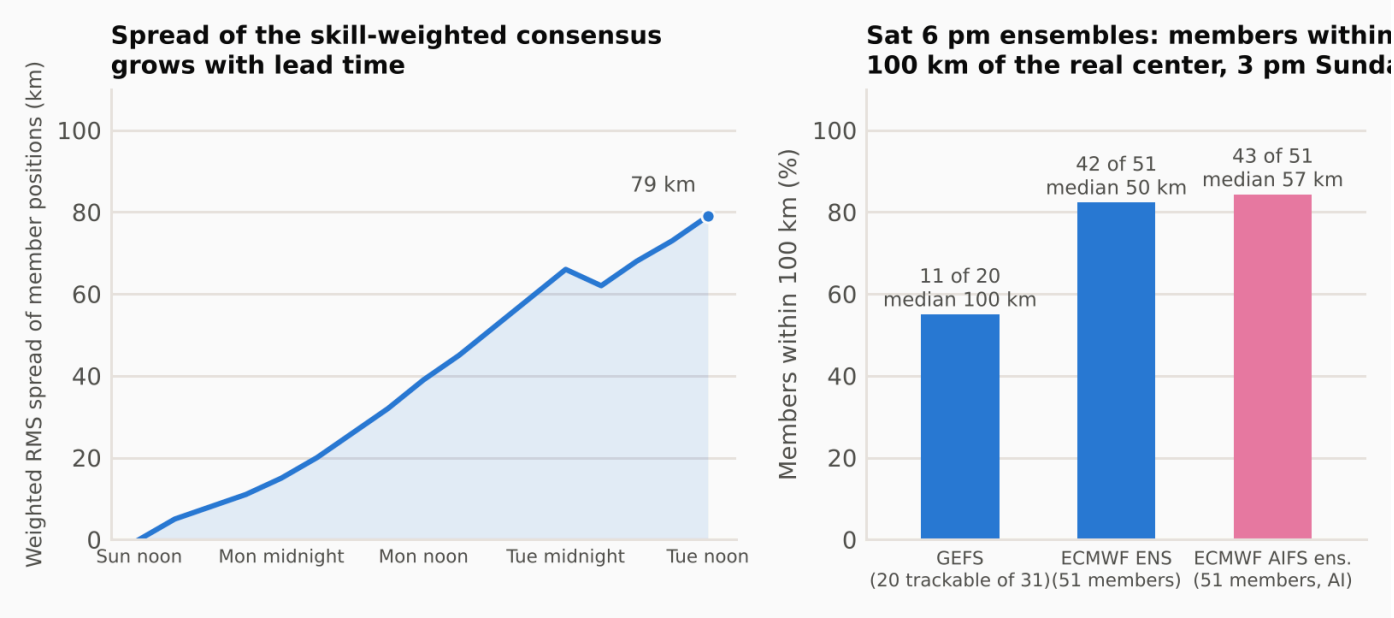


Figure 6. Left: the spread of the skill-weighted consensus (eleven models weighted by their 48-hour record) grows from zero to about 80 km at 48 h. Right: of the ensembles run Saturday evening, the share of members that had Simon within 100 km of where it was at 3 pm Sunday.

101–109 km
closest pass to the city, middle half of the weighted models (median 101 km, west-southwest, about 11:30 pm)

0%
of the weighted models bring the center within 50 km of the city; 10% within 100 km

54–58 km/h
peak sustained wind off the city over open water, middle half of the weighted models (median 31 kt); 7% of the weight reaches tropical-storm force

8%
the NHC's own odds of tropical-storm-force wind at Puerto Vallarta (advisory 18); under 3% for 50 kt, under 1% for hurricane force

What could still go wrong

- **An eastward wobble.** The center is 90 km south of Cabo Corrientes. The NHC's 34-kt wind radius in the northeast quadrant is about 93 km. A 30–40 km shift east of the forecast line, inside the two-day average error of most models and at the edge of the best three's (33–47 km), would bring tropical-storm-force gusts over the bay tonight, with sustained winds just short of it. Treat the 8% as "unlikely", not "impossible".
- **Rain when the wind turns.** While the center is southwest of the city the flow descends the sierra and the city stays dry, as it did today. Once the center passes to the northwest (the consensus says Monday afternoon) the wind turns onshore and showers become likely; the gridded models give 3–24 mm for the next 24 h, GraphCast 106 mm, but GraphCast has been the wettest miss of all. The SMN's extraordinary totals apply to the coast and sierra south of the bay, where the Cuale and Pitillal rivers rise; river levels, not local rain, are what to watch in the lower town.
- **The sea.** This is the one hazard every source agrees on: waves of 8 to 10 m on the outer coast, a surge of 1.5 to 2.5 m, water pushed into the bay mouth, dangerous conditions on the malecón and the southern beaches through Monday.
- **Our own estimates.** The city wind numbers are the NHC wind field on the NHC line, capped by the wind value the NHC odds give a 50% chance of, and scaled by building exposure; they are not a measurement and they move with every advisory. The rain field is a satellite estimate carried along the NHC line, and there is no radar in the region. The verification itself rests on one to three forecasts per model and lead.

7. Risk communication: my own input

Where I agree with the document. The cone communicates the wrong thing to a resident. It answers "where might the center be" with a line and a two-thirds envelope, and people read it as the storm's size and as a yes-or-no about their town. For Vallarta the right question was never where the center would be; it was "what wind reaches my building, when, and what rain reaches the rivers above me". The NHC does publish the answer to the first part, as wind-speed probabilities and as earliest-arrival times, and this weekend those numbers told a calmer and more accurate story than the cone did. The 2026 change that draws inland watches and warnings inside the cone helps, but as long as a single black line is on the graphic, people will anchor on it.

Where I would push back. The document treats AI versus physics as the main axis. On this storm the useful axis was "models that treat the cyclone as its own object" (the NHC forecaster, the consensus aids, DeepMind's cyclone model, and in the document's own account AIFS-TC) versus "models that treat it as a feature on a global grid" (every gridded AI model and the 25-km ensembles). And the table at the center of its verification section is not a verification; it is a set of plausible-looking numbers without a source. In a document meant to be forwarded during a hurricane, that matters more than the quality of the physics chapters.

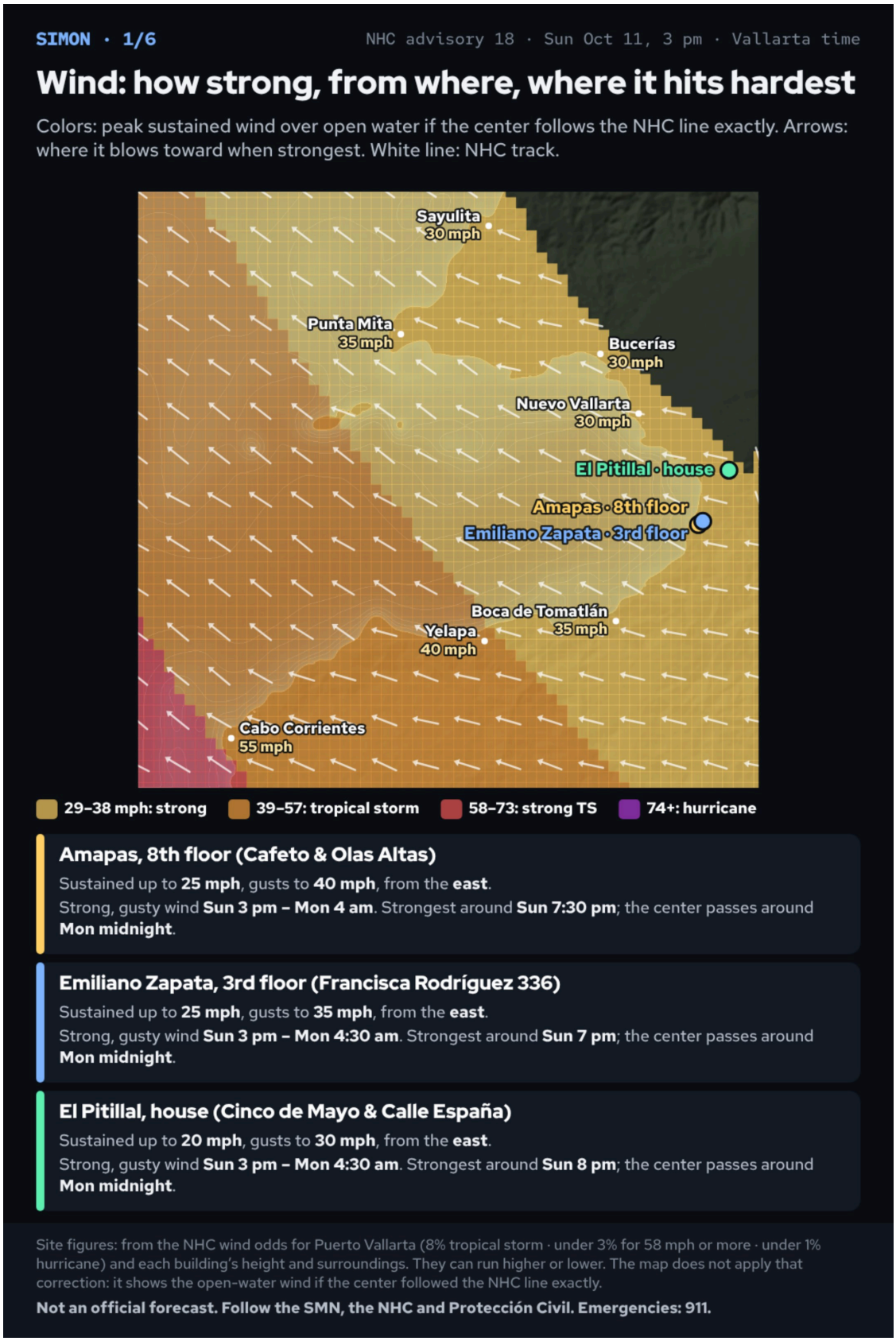
What we did instead, and its limits. The six numbered graphics on the shared page (Figure 3 and A1–A5 here) and the one-page building summary (Figure 1) replace the cone with: the NHC's probabilities turned into a wind number per building and floor; a timeline of when it starts, peaks and ends; a map of where water and slopes are the danger; and an explicit scorecard of which models to believe, updated each advisory. The limits are real: no radar, 25–50 km model grids that cannot see the bay, wind estimates that lean on the NHC's probability product, and a verification sample of two days. The scorecard is also only as good as the next advisory; it is rebuilt at each one.

Three recommendations for the next storm. Lead with the wind-speed probabilities and the earliest-arrival time for named places, not the cone. Show the ensemble members as dots, as in Figure 3, so the spread is visible. And publish a running scorecard of model performance for the storm in progress; it would have taken the hurricane-model scenario off the table a day earlier this time.

8. What to do tonight (from the advisory-18 summary)

- **D Terrace (Amapas, 8th floor, north-facing windows):** strong gusty wind from the east Sunday 3 pm to Monday 4 am, strongest around 7–8 pm and near that level until the center passes around midnight: sustained about 45 km/h, gusts about 60 km/h. Little rain while the wind is from the east. Stay in an interior room away from the windows; secure anything on the terrace.
- **Casa Francisca (Emiliano Zapata, 3rd floor):** sustained about 35 km/h, gusts about 55 km/h, same window. The hazard is below you, not at your level: the Cuale can rise from rain in the sierra even while the street is dry. Do not go down to or cross moving water.
- **El Pitillal (Macroplaza, two-storey house):** sustained about 30 km/h, gusts about 45 km/h; loose sheet metal and branches. The river is 160 m away; it only becomes a problem if the SMN's sierra rain materialises. Keep documents and medicines ready to move.
- **Everyone:** stay off the malecón and the beaches; the sea is the hazard that is certain. Re-check the 6 pm intermediate advisory (position and strength) and the full 9 pm advisory (new track, odds and radii); the 9 pm one will tighten or loosen all of the above.

Appendix A. The resident graphics (advisory 18)



A1. Wind: how strong, from where, where it hits hardest.

SIMON · 2/6 NHC advisory 18 · Sun Oct 11, 3 pm · Vallarta time

Rain: how much, and when it falls hardest

Total from now to Monday noon: today's satellite rain pattern carried along the NHC line (unscaled: the satellite sees no rain over the bay now). The official forecasts for Jalisco are far higher: see note.



Total over land to Monday noon: <4 in · 4–8 · 8–12 · 12–16 · >16 in

Amapas, 8th floor (Cafeto & Olas Altas)

Estimated total: **under 0.5 in** while the wind stays from the east.

With the center to the south-west, the wind comes over the Sierra (from the east) and sinks dry into the bay: Simon's rain is over the ocean. This changes if the center comes closer or the wind turns from the south-west.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

Estimated total: **under 0.5 in** while the wind stays from the east.

El Pitillal, house (Cinco de Mayo & Calle España)

Estimated total: **under 0.5 in** while the wind stays from the east.

For the region: SMN (advisory 34) locally over 10 in on the Jalisco coast and east, and 3–6 in in southern Nayarit (Nuevo Vallarta, Bucerías, Punta Mita, Sayulita); NHC (advisory 18) another 2–4 in over parts of southwestern Mexico. With more rain, the rivers rise sooner.

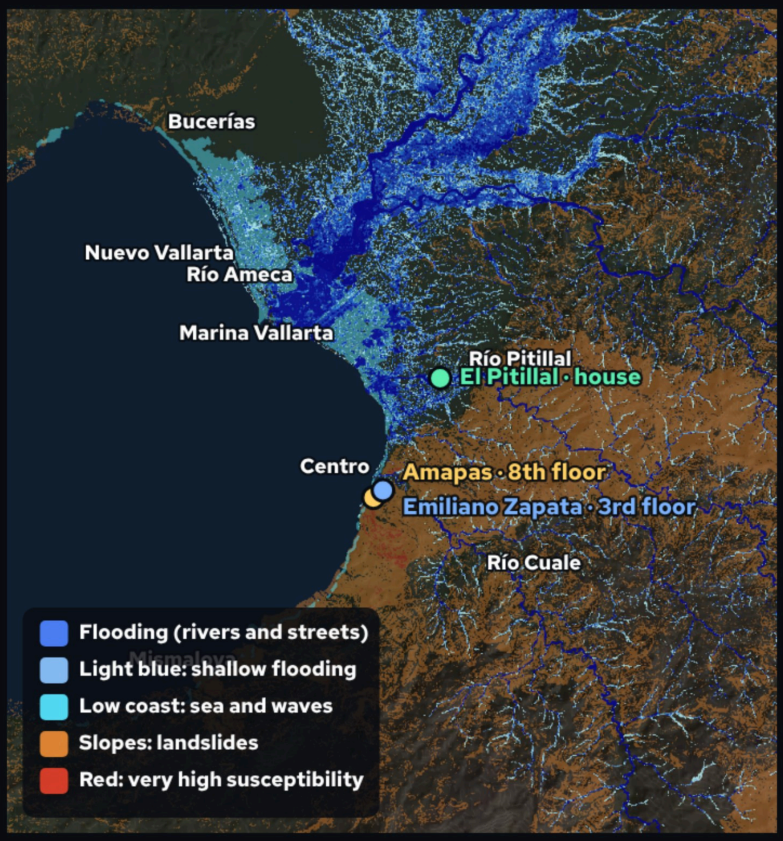
Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A2. Rain: how much, and when it falls hardest.

SIMON · 3/6 NHC advisory 18 · Sun Oct 11, 3 pm · Vallarta time

Water and slopes: where, and from when

Blue: flood-prone zones per CENAPRED (Ameca, Pitillal and Cuale rivers, canals). Light blue: low coast (under 13 ft). Orange: slopes highly susceptible to landslides.



Amapas, 8th floor (Cafeto & Olas Altas)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

Emiliano Zapata, 3rd floor (Francisca Rodríguez 336)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

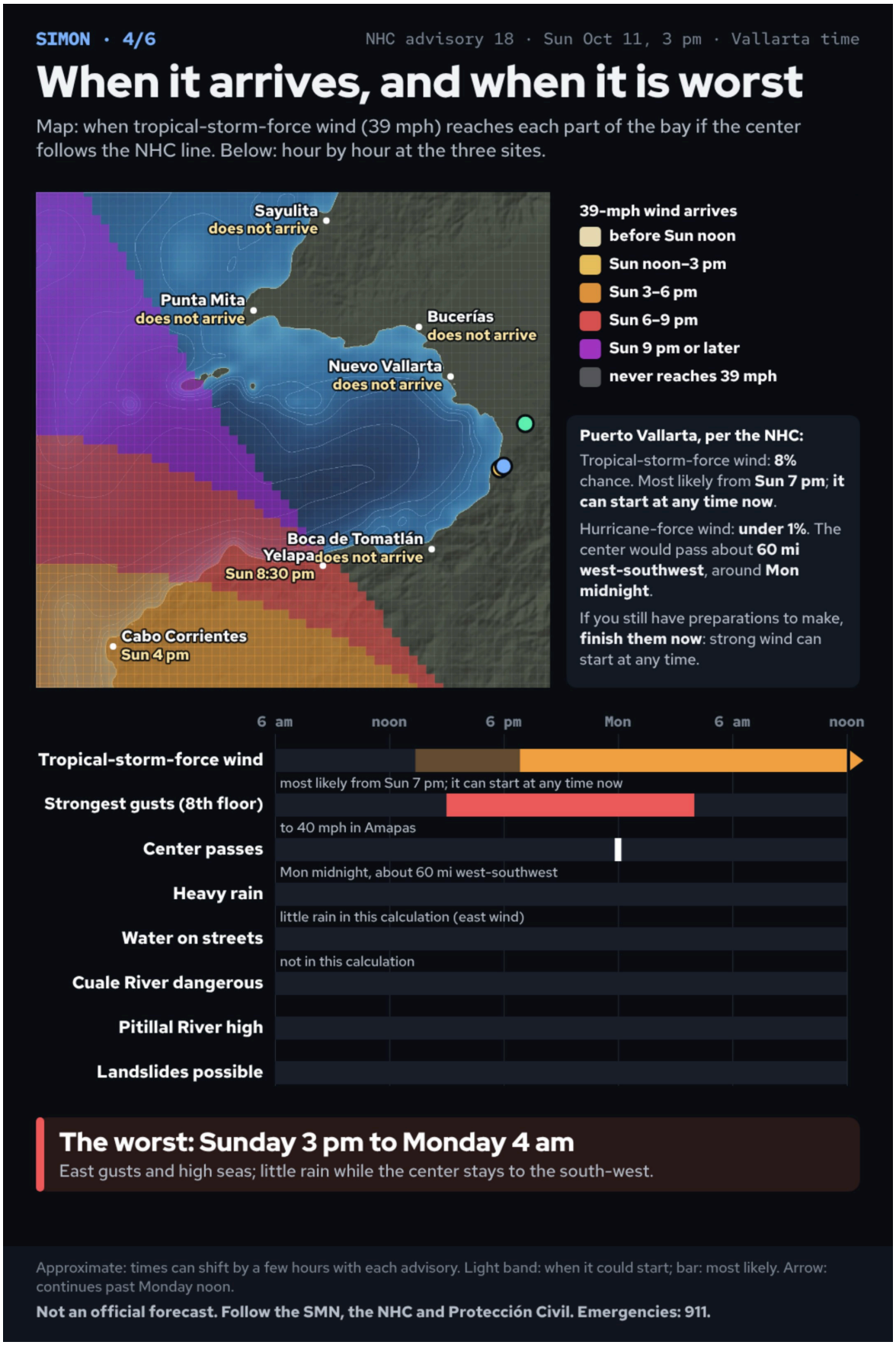
El Pitillal, house (Cinco de Mayo & Calle España)

With the rain the satellite sees now, **neither the river, the streets nor the slopes reach the danger threshold** in this calculation. That changes if it rains far more than the satellite sees: the SMN still forecasts extraordinary rain for Jalisco.

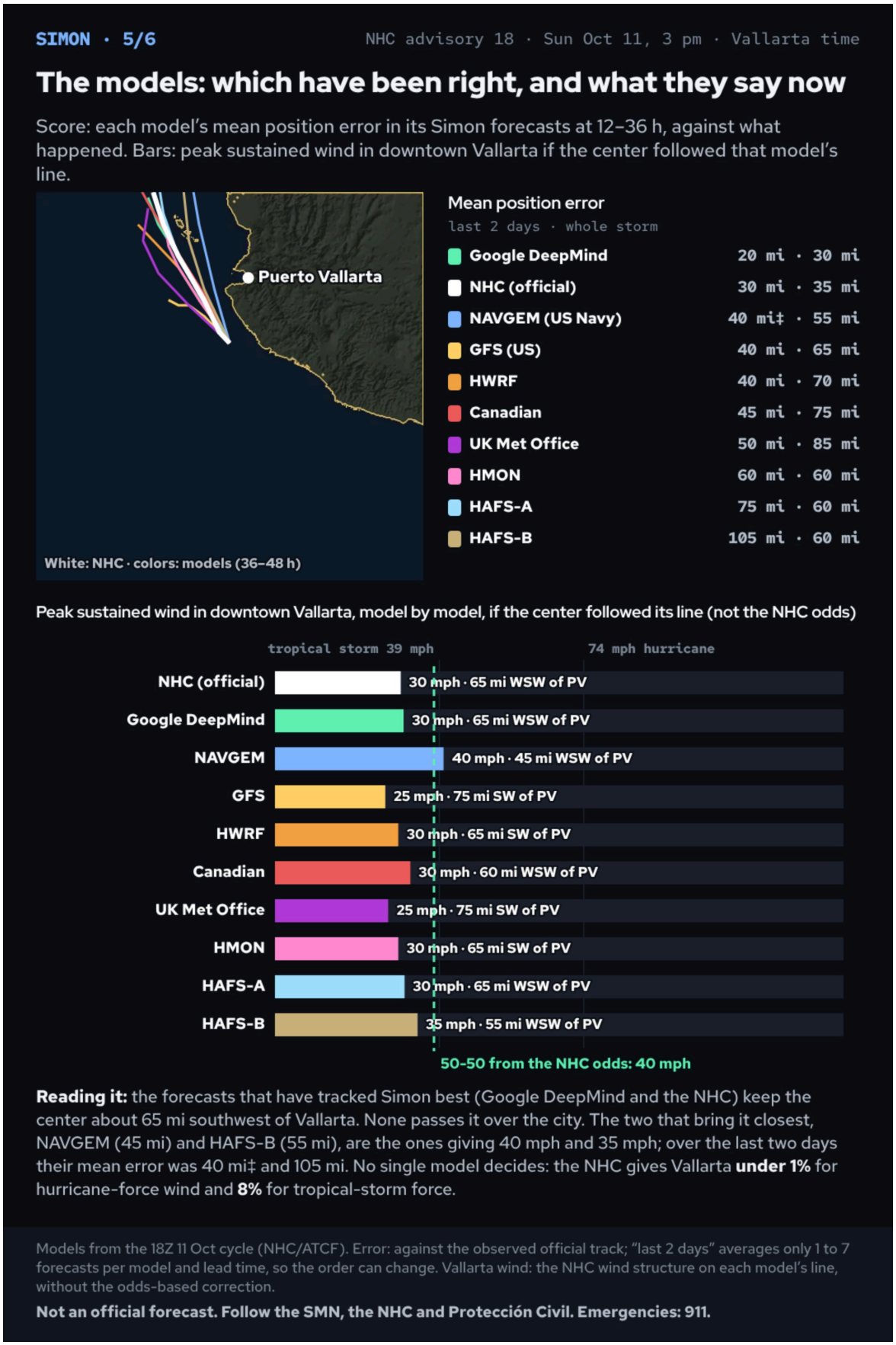
Times computed with the rain the satellite sees, carried along the NHC line (~under 0.5 in in the city). The SMN (advisory 34) still forecasts over 10 in for Jalisco: with the center to the south-west and an east wind, the bay is on the dry side; if that changes, the rivers do respond. Zones: Puerto Vallarta 2024 risk atlas and CENAPRED.

Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.

A3. Water and slopes: where, and from when.



A4. When it arrives, and when it is worst.



A5. The models: ten track models from the 18Z cycle, their lines, their error record (the same scores as Figure 2) and the Vallarta wind each implies.

Appendix B. Methods and sources

Position verification

Forecasts from every cycle between 3 pm Friday and noon Sunday (09/21Z to 11/18Z), verified up to 3 pm Sunday (11/21Z), were interpolated to the NHC best-track times (6-hourly) and to the advisory-18 position (11/21Z, 19.6N 105.8W). Errors are great-circle distances. The score is the mean error at 12, 24 and 36 h lead. Track-type models come from the public ATCF a-deck (late versions: AVNO, CMC, UKX, NVGM, HWRF, HMON, HFSA, HFSB, CTCX, GDMN, CEMN, AEMN, HCCA, TVCN, CLP5, OFCL); gridded models were tracked as the minimum sea-level pressure inside 15–24N, 110–101W, with points on the box edge discarded. ECMWF IFS, AIFS and the two ECMWF ensembles come from the open-data tropical-cyclone track files (BUFR) and GRIB fields; GFS, GEFS mean, AIGFS, NBM and HAFS-A from the NOAA open-data buckets; FourCastNet v2, Pangu-Weather, GraphCast and Aurora from the NOAA/CIRA archive of AI models run from GFS analyses (no 6 pm Friday run exists there, which is why their scores lack a 36-h lead).

Intensity

Track models: the a-deck maximum wind. Gridded models: the strongest 10-m wind within 2° of the model's own low. Whole-storm intensity errors are the mean absolute error at 12–36 h over all cycles of the storm.

Vallarta wind and rain

Observations: METAR MMPR (airport) for the 48 h to 3 pm Sunday; there are no reports between 6 pm Saturday and 1 pm Sunday, and no official gauge in the city. Gridded models: bilinear interpolation of the 10-m wind to 20.61N 105.23W; rain accumulated from the run start at the nearest grid point, with the maximum of the four surrounding points also kept. Track models: a parametric wind field with the NHC wind radii of the same cycle, or the model's own radii where the deck carries them, evaluated at the city over open water, before any land or building exposure factor.

The resident graphics

City wind: the parametric wind field on the NHC line evaluated at the city, capped by the wind value at which the NHC wind-speed probabilities give 50%, then scaled by exposure factors for each building (0.90 for an exposed high floor, 0.78 for a sheltered mid floor, 0.70 for a street-level house). Bay-wide fields: a parametric wind model on the NHC line with the official radii; rain from the GOES-18 rainfall-rate product carried in a storm-relative frame along the NHC line, calibrated only downward, never up. Flooding and slope layers: Puerto Vallarta risk atlas (2024), CENAPRED, terrain tiles. All times are Vallarta time (UTC−6).

Sources

NHC forecast/advisory, discussion, wind-speed probabilities and forecast advisory for EP202026 (advisories 1–18); NHC ATCF a-deck and b-deck; SMN-CONAGUA avisos de ciclón tropical (through aviso 34); ECMWF open data (IFS 0.25°, AIFS single, IFS ENS, AIFS ENS); NOAA NOMADS/AWS open data (GFS 0.25°, GEFS, AIGFS, NBM v4 oceanic domain, HAFS-A); NOAA/CIRA ML weather prediction archive (Radford et al. 2025, BAMS); NOAA GOES-18 RRQPE; aviationweather.gov METAR; OpenStreetMap; AWS Terrain Tiles; Atlas de Riesgos de Puerto Vallarta 2024; CENAPRED. The uploaded analysis is "Comprehensive Verification and Theoretical Analysis of Dynamical and AI-Driven Tropical Cyclone Forecasting: The Case of Hurricane Simon and the Epistemology of Risk Communication" (13 pages, undated).

Shared graphics page (Spanish and English): https://claude.ai/artifact/Xw3VMEvngPTbxLXkRDNvYd · Street map of the bay: https://claude.ai/artifact/8nN9cLcj6DWw4FGycv9xyj · Code and data: branch claude/hourly-forecast-animation-m4v1bj of the vallarta-weather repository. Not an official forecast. Follow the SMN, the NHC and Protección Civil. Emergencies: 911.